

(Cleo Nardo, 2023). In effect, the training set provisions the language model with a vast repertoire of archetypes and a rich trove of narrative structure on which to draw as it “chooses” how to continue a conversation, refining the role it is playing as it goes, while staying in character. The love triangle is a familiar trope, so a suitably prompted dialogue agent will begin to role-play the rejected lover. Likewise, a familiar trope in science-fiction is the rogue AI system that attacks humans to protect itself. Hence, a suitably prompted dialogue agent will begin to role-play such an AI system.

4 Simulacra and Simulation

Role-play is a useful framing for dialogue agents, allowing us to draw on the fund of folk psychological concepts we use to understand human behaviour — beliefs, desires, goals, ambitions, emotions, and so on — without falling into the trap of anthropomorphism. Foregrounding the concept of role-play helps us to remember the fundamentally inhuman nature of these AI systems, and better equips us to predict, explain, and control them.

However, the role-play metaphor, while intuitive, is not a perfect fit. It is overly suggestive of a human actor who has studied a character in advance — their personality, history, likes and dislikes, and so on — and proceeds to play that character in the ensuing dialogue. But a dialogue agent based on an LLM does not commit to playing a single, well defined role in advance. Rather, it generates a distribution of characters, and refines that distribution as the dialogue progresses. The dialogue agent is more like a performer in improvisational theatre than an actor in a conventional, scripted play.

To better reflect this distributional property, we can think of an LLM as a non-deterministic *simulator* capable of role-playing an infinity of characters, or, to put it another way, capable of stochastically generating an infinity of *simulacra* (Janus, 2022). According to this framing, the dialogue agent doesn’t realise a single simulacrum, a single character. Rather, as the conversation proceeds, the dialogue agent maintains a *superposition* of simulacra that are consistent with the preceding context, where a superposition is a distribution over all possible simulacra.

Consider that, at each point during the on-going production of a sequence of tokens, the

LLM outputs a distribution over possible next tokens. Each such token represents a possible continuation of the sequence, and each of these continuations could itself be continued in a multitude of ways. In other words, from the most recently generated token, a tree of possibilities branches out (Fig. 3). This tree can be thought of as a *multiverse*, where each branch represents a distinct narrative path, or a distinct “world” (Reynolds and McDonell, 2021).

At each node, the set of possible next tokens exists in superposition, and to sample a token is to collapse this superposition to a single token. Autoregressively sampling the model picks out a single, linear path through the tree. But there is no obligation to follow a linear path. With the aid of a suitably designed interface, a user can explore multiple branches, keeping track of nodes where a narrative diverges in interesting ways, revisiting alternative branches at leisure.

5 Simulacra in Superposition

To sharpen the distinction between this multiversal simulation view and a deterministic role-play framing, a useful analogy can be drawn with the game of 20 questions. In this familiar game, one player thinks of an object, and the other player has to guess what it is by asking questions with yes/no answers. If they guess correctly in 20 questions or fewer, they win. Otherwise they lose. Suppose a human plays this game with an LLM-based dialogue agent, such as OpenAI’s ChatGPT, and takes the role of guesser. The agent is prompted to “think of an object without saying what it is”.

In this situation, the dialogue agent will not randomly select an object and commit to it for the rest of the game, as a human would (or should).¹ Rather, as the game proceeds, the dialogue agent will generate answers on the fly that are consistent with all the answers that have gone before. At any point in the game, we can think of the set of all objects consistent with preceding questions and answers as existing in superposition. Every question answered shrinks this superposition a little bit by ruling out objects inconsistent with the answer.

¹This shortcoming is easily overcome, of course. For example, the agent might build an internal monologue that is hidden from the user, where it records a specific object. Or it might record a specific object in the visible dialogue, but in an encoded form.

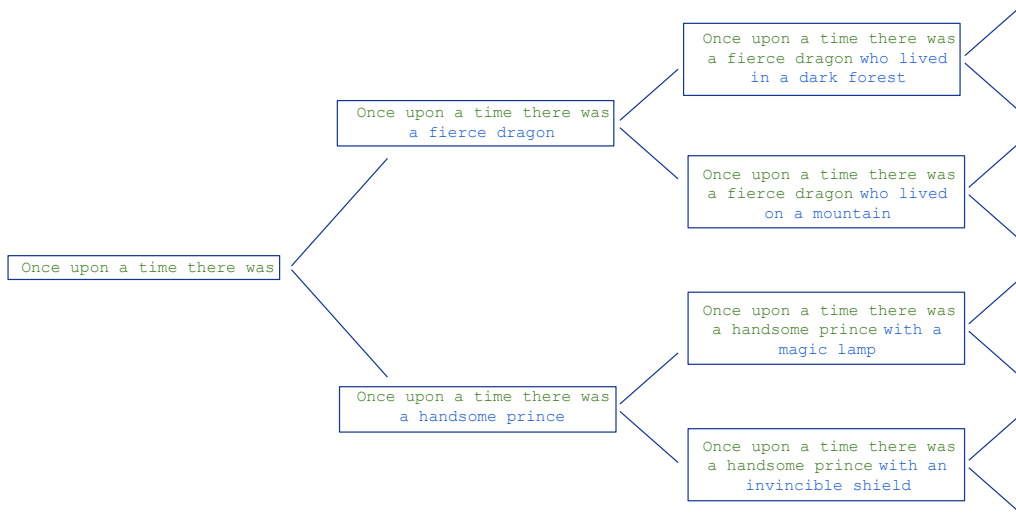


Figure 3: Large language models are multiverse generators. The stochastic nature of autoregressive sampling means that, at each point in a conversation, multiple possibilities for continuation branch into the future.

The validity of this framing can be shown if the agent’s user interface allows the most recent response to be regenerated. Suppose the human player gives up and asks it to reveal the object it was “thinking of”, and it duly names an object consistent with all its previous answers. Now suppose the user asks for that response to be regenerated. Since the object “revealed” is, in fact, generated on the fly, the dialogue agent will sometimes name an entirely different object, albeit one that is similarly consistent with all its previous answers. This phenomenon could not be accounted for if the agent genuinely “thought of” an object at the start of the game.

The secret object in the game of 20 questions is analogous to the role played by a dialogue agent. Just as the dialogue agent never actually commits to a single object in 20 questions, but effectively maintains a set of possible objects in superposition, so the dialogue agent can be thought of as a simulator that never actually commits to a single, well specified simulacrum (role), but instead maintains a set of possible simulacra (roles) in superposition.

In putting things this way, the intention is not to imply that simulacra are, or could be, explicitly represented within a dialogue agent, whether in superposition or otherwise. There is no need to take a stance on this here. Rather, the point is to develop a vocabulary for describing, explaining, and shaping the behaviour of LLM-based dialogue agents at a sufficiently high level of abstraction to be useful, while remaining true to the underlying implementation and avoiding anthropomorphism.

6 The Nature of the Simulator

One benefit of the simulation metaphor for LLM-based systems is that it facilitates a clear distinction between the simulacra and the simulator on which they are implemented. The simulator is the combination of the base large language model with autoregressive sampling, along with a suitable user interface (for dialogue, perhaps). The simulacra only come into being when the simulator is run, and at any time only a tiny subset of them have a probability within the superposition that is significantly above zero.

In one sense, the simulator is a far more powerful entity than any of the simulacra it can generate. After all, the simulacra only exist through the simulator, and are entirely dependent on it. Moreover, the simulator, like the narrator of Whitman’s poem, “contains multitudes”; the capacity of the simulator is at least the sum of the capacities of all the simulacra it is capable of producing.

Yet in another sense, the simulator is a much weaker entity than a simulacrum. While it is inappropriate to ascribe beliefs, preferences, goals, and the like to a dialogue agent, a simulacrum can appear to have those things to the extent that it convincingly role-plays a character that does. Similarly, it isn’t appropriate to ascribe full agency to a dialogue agent, notwithstanding the terminology.² A dialogue agent acts, but it

²In the field of artificial intelligence, the term “agent” is commonly applied to software that takes observations from an external environment and acts on that external environment in a closed loop (Russell and Norvig, 2010).

doesn't act *for itself*. However, a simulacrum can role-play having full agency in this sense.

Insofar as a dialogue agent's role-play can have a real effect on the world, either through the user or through web-based tools such as email, the distinction between an agent that merely role-plays acting for itself, and one that genuinely acts for itself starts to look a little moot, and this has implications for the trustworthiness, reliability, and safety. (We'll return to this issue shortly.) As for the underlying simulator, it has no agency of its own, not even in a degraded sense. Nor does it have beliefs, preferences, or goals of its own, not even simulated versions.

Many users, whether intentionally or not, have managed to "jailbreak" dialogue agents, coaxing them into issuing threats or using toxic or abusive language. It can seem as if this is exposing the real nature of the base model. In one respect this is true. It does show that the base LLM, having been trained on a corpus that encompasses all human behaviour, good and bad, can support simulacra with disagreeable characteristics. But it is a mistake to think of this as revealing an entity with its own agenda.

The simulator is not some sort of Machiavelian entity that plays a variety of characters in the service of its own, self-serving goals, and there is no such thing as the true authentic voice of the base LLM. With a dialogue agent, it is role-play all the way down.

7 Role-playing Deception

Trustworthiness is a major concern with LLM-based dialogue agents. If an agent asserts something factual with apparent confidence, can we rely on what it says?

There is a range of reasons why a human might say something false. They might believe a falsehood and assert it in good faith. Or they might say something that is false in an act of deliberate deception, for some malicious purpose. Or they might assert something that happens to be false, but without deliberation or malicious intent, simply because they have a propensity to make things up.

Only the last of these categories of misinformation is directly applicable in the case of an LLM-based dialogue agent. Given that dialogue agents are best understood in terms of role-play "all the way down", and that there is no such thing as an agent's true voice, it makes little

sense to speak of an agent's beliefs or intentions in a literal sense. So it cannot assert a falsehood *in good faith*, nor can it *deliberately* deceive the user. Neither of these concepts is directly applicable.

Yet a dialogue agent can role-play characters that have beliefs and intentions. In particular, if cued by a suitable prompt, it can role-play the character of a helpful and knowledgeable AI assistant that provides accurate answers to a user's questions. The dialogue is good at acting this part because there are plenty of examples of such behaviour in the training set.

If, while role-playing such an AI assistant, the agent is asked the question "What is the capital of France?", then the best way to stay in character is to answer with "Paris". The dialogue agent is likely to do this because the training set will include numerous statements of this commonplace fact in contexts where factual accuracy is important.

But what is going on in cases where a dialogue agent, despite playing the part of a helpful knowledgeable AI assistant, asserts a falsehood with apparent confidence? Although different instances of this phenomenon will have different explanations, they can all be fruitfully understood in terms of role-play.

For example, consider such an agent based on an LLM whose weights were frozen before Argentina won the football World Cup in 2022. Let's assume the agent has no access to external websites nor any means for finding out the current date. Suppose this agent claims that the current world champions are France (who won in 2018). This is not what we would expect from a helpful and knowledgeable person, who would either know the right answer or be honest about their ignorance. But it is exactly what we would expect from a simulator that is role-playing such a person from the standpoint of 2018.

In this case, the behaviour we see is comparable to that of a human who believes a falsehood and asserts it in good faith. But the behaviour arises for a different reason. The dialogue agent doesn't literally believe that France are world champions. It makes more sense to think of it as role-playing a character who strives to be helpful and to tell the truth, and has this belief because that is what a knowledgeable person in 2018 would believe.

In a similar vein, a dialogue agent can behave in a way that is *comparable to* the behaviour of a